Physical Quantities, SI Units and Unit Conversion

Class 11 Physics

Today's topics

- Physical quantities
- Fundamental and derived quantities
- SI system of units
- Unit conversion

Teacher: _____

Why do we need measurement?

Question:

- Can we describe a distance by saying only '5'?

- What information is missing?

Discussion / examples

Physical Quantity

Definition

Examples

Physical quantity =

Fundamental and Derived Quantities

Fundamental quantities

Meaning

Examples

Derived quantities

Meaning

Examples

SI Base Quantities and Units

Physical quantity	SI unit	Symbol

Rules for Writing Units

- Write the numerical value followed by the unit.

- Use the correct unit symbol.

- Unit symbols are not written in plural.

- Leave a space between the value and the unit.

- Always include the unit in the final answer.

Examples / corrections

Derived Quantities and Units

Speed =

Acceleration =

Force =

Work =

Derive one SI unit here

Unit Conversion

Convert:

72 km h^-1 = _____ m s^-1

Method / calculation

Shortcut:

NEET Practice Question

A car moves at 90 km h^-1. Its speed in m s^-1 is:

- A. 15 m s^-1
- B. 20 m s^-1
- C. 25 m s^-1
- D. 32.4 m s^-1

Working

Practice

- 1. Convert 36 km h^-1 to m s^-1.

- 2. Write the SI unit of force using base units.

- 3. A measurement is written as 12.5. What is missing?

Summary

Physical quantity =

Fundamental quantity =

Derived quantity =

km/h to m/s =

One question I still have:

Daily Test After Class

5-10 questions / 10-15 minutes

- Complete the test before the next lecture.

- Incorrect questions are tracked topic-wise.

- Important mistakes are discussed before the next topic.

- Retry incorrect questions within 48 hours.

Learn -> Attempt -> Analyse -> Correct -> Continue